

Moduli spaces of pointed genus one curves and logarithms of algebraic numbers

Tiago J. Fonseca

Unicamp

VI Congresso Brasileiro de Jovens Pesquisadoras e
Pesquisadores em Matemática Pura, Aplicada e Estatística
Teresina, May 2026

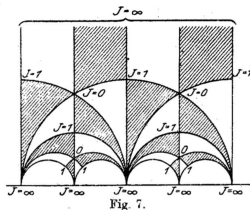
- ▶ Joint work with **Francis Brown**.
- ▶ Preprint [arXiv:2508.04844]:
Single-valued periods of meromorphic modular forms and a motivic interpretation of the Gross–Zagier conjecture
- ▶ Goals of the talk:
 - ▶ Recall the GZ log-algebraicity conjecture.
Note: Li '23, Bruinier-Li-Yang '25: analytic proof.
 - ▶ Relate it to the moduli spaces $M_{1,n}$.
 - ▶ Discuss geometric and motivic aspects.

The Gross–Zagier conjecture

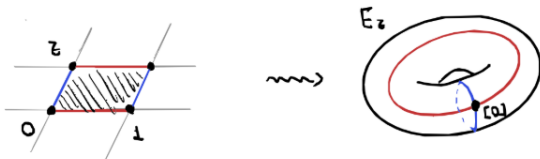
- $\mathbb{H} = \{z \in \mathbb{C} : \text{Im}(z) > 0\}$ upper-half plane; $SL_2(\mathbb{Z})$ -action:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az + b}{cz + d}.$$

- Klein's $J : \mathbb{H} \rightarrow \mathbb{C}$ inducing $SL_2(\mathbb{Z}) \backslash \mathbb{H} \xrightarrow{\sim} \mathbb{C}$.



- Moduli: $SL_2(\mathbb{Z}) \backslash \mathbb{H} \cong M_{1,1}$ via $z \mapsto (E_z = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}z), [0])$.



Theorem (Kronecker)

If $z \in \mathbb{H}$ is a quadratic number, then $J(z) \in \overline{\mathbb{Q}}$.

- ▶ Example (Weber):

$$J(\sqrt{-14}) = 6^{-3} \left(323 + 228\sqrt{2} + (231 + 161\sqrt{2})\sqrt{2\sqrt{2} - 1} \right)^3$$

- ▶ Geometric interpretation: an elliptic curve with too many endomorphisms (i.e., $\text{End}(E) \supsetneq \mathbb{Z}$) is defined over $\overline{\mathbb{Q}}$.
- ▶ We say $z \in \mathbb{H}$ is a **CM point**.

*"The theory of **complex multiplication**, which forms a powerful link between number theory and analysis, is not only the most beautiful part of mathematics but also of all science." D. Hilbert*

Heegner points and derivatives of L -series

Benedict H. Gross¹ and Don B. Zagier²

¹ Department of Mathematics, Brown University, Providence, RI 02912, USA

² Department of Mathematics, University of Maryland, College Park, MD 20742, USA and
Max-Planck-Institut für Mathematik, Gottfried-Claren-Strasse 26, D-5300 Bonn 3,
Federal Republic of Germany

- ▶ Key case of BSD conjecture.
- ▶ Computation involves **higher Green's functions**:

$$G_s(z, w) = -2 \sum_{\gamma \in SL_2(\mathbb{Z})} Q_{s-1} \left(1 + \frac{|z - \gamma w|^2}{2\operatorname{Im}(z)\operatorname{Im}(\gamma w)} \right)$$

where $z, w \in \mathbb{H}$ not in the same orbit, $s \geq 1$.

- ▶ 'Higher' versions of $G_1(z, w) = \log |J(z) - J(w)|^2$, but **unclear geometric interpretation**.

When $s \geq 1$ **is an integer**, G_s has remarkable arithmetic properties:

- ▶ **GZ conjecture** for $s \in \{1, 2, 3, 4, 5, 7\}$: if z and w are CM of discriminants d_z and d_w , then

$$G_s(z, w) = (d_z d_w)^{\frac{s-1}{2}} \log |\alpha|, \quad \text{for some } \alpha \in \overline{\mathbb{Q}}$$

More complicated version for other s .

- ▶ For $G_1(z, w) = \log |J(z) - J(w)|^2$, it follows from Kronecker.
- ▶ Example (Mellit):

$$G_2\left(\frac{-1+i\sqrt{7}}{2}, i\right) = \frac{8}{\sqrt{7}} \log(8 - 3\sqrt{7}).$$

- ▶ Previous work: Gross–Kohnen–Zagier '87, Zhang '97, Mellit '08, Viazovska '11, ... , Li '23, Bruinier–Li–Yang '25.

Our goal: relate the GZ conjecture to Algebraic Geometry.

Differential forms on $M_{1,n}$

Example (s=1)

- $G_1(z, w) = \log |J(z) - J(w)|^2$ as a definite integral:

$$G_1(z, w) = \int_1^{J(z)} \frac{dx}{x - J(w)} + \overline{\int_1^{J(z)} \frac{dx}{x - J(w)}}$$

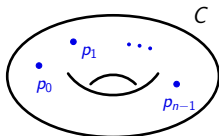
- As a primitive:

$$d_z G_1(z, w) = \underbrace{\frac{J'(z)dz}{J(z) - J(w)}}_{\nu} + \overline{\frac{J'(z)dz}{J(z) - J(w)}} = \nu + \bar{\nu}$$

- ν is a '1-form of the 3rd kind' on $M_{1,1}$ with poles at $J(w)$.

Question: do $G_s(z, w)$ admit a similar description for $s > 1$?

- ▶ $M_{1,n}$ = moduli of genus 1 curves with n marked points:



$\overline{M}_{1,n}$ = Deligne–Mumford compactification (stable curves).

- ▶ Action of S_n : permute the marked points.
- ▶ “Uniformisation”:

$$\Gamma_{1,n} \backslash U_{1,n} \xrightarrow{\sim} M_{1,n}$$

$$(z, u_1, \dots, u_{n-1}) \mapsto (E_z = \mathbb{C}/(\mathbb{Z} + z\mathbb{Z}), [0], [u_1], \dots, [u_{n-1}])$$

where

$$U_{1,n} = \{(z, u_1, \dots, u_{n-1}) \in \mathbb{H} \times \mathbb{C}^{n-1} : u_i \notin \mathbb{Z} + \mathbb{Z}z, u_i - u_j \notin \mathbb{Z} + \mathbb{Z}z\}$$

$$\Gamma_{1,n} = SL_2(\mathbb{Z}) \ltimes (\mathbb{Z}^2)^{n-1}$$

- For $n \geq 3$ odd and $p + q = n - 1$, $p, q \geq 0$, let

$$\psi^{p,q}(z, w) = \sum_{\gamma \in SL_2(\mathbb{Z})} \frac{1}{(cz + d)^{n+1}} \frac{w - \bar{w}}{(\gamma z - w)^{p+1}(\gamma z - \bar{w})^{q+1}}$$

and set

$$\nu^{p,q} = \psi^{p,q}(z, w) dz \wedge du_1 \wedge \cdots \wedge du_{n-1}$$

- Let D_w be the fibre of $J(w) = (E_w, [0])$ under

$$\pi : \overline{M}_{1,n} \rightarrow \overline{M}_{1,1}, \quad (C, p_0, \dots, p_{n-1}) \mapsto (C, p_0)$$

Theorem (Brown–F.)

The above $\nu^{p,q}$ define algebraic n -forms on $\overline{M}_{1,n} \setminus D_w$ whose cohomology classes are S_n -alternating and satisfy

$$H^n(\overline{M}_{1,n} \setminus D_w)_{sgn} = H^n(\overline{M}_{1,n})_{sgn} \oplus \bigoplus_{p+q=n-1} \mathbb{C} \cdot [\nu^{p,q}]$$

Theorem (Brown–F.)

There are unique real-analytic $(n-1)$ -forms $g^{p,q}$ on $\overline{M}_{1,n} \setminus D_w$ such that

$$dg^{p,q} = \nu^{p,q} + \overline{\nu^{q,p}}, \quad g^{p,q} = \sum_{r+s=n-1} G_{r,s}^{p,q}(z, w) \omega^r \overline{\omega}^s$$

where

$$\omega_i = du_i - \frac{u_i - \overline{u}_i}{z - \overline{z}} dz$$

and, for $r + s = n - 1$, set

$$\omega^r \overline{\omega}^s = \frac{1}{(n-1)!} \sum_{\sigma \in S_{n-1}} \operatorname{sgn}(\sigma) \omega_{\sigma(1)} \wedge \cdots \wedge \omega_{\sigma(r)} \wedge \overline{\omega}_{\sigma(r+1)} \wedge \cdots \wedge \overline{\omega}_{\sigma(n-1)}$$

Moreover,

$$G_{s-1,s-1}^{s-1,s-1}(z, w) = (-1)^{s-1} \binom{2s-2}{s-1} \frac{G_s(z, w)}{(z - \overline{z})^{s-1} (w - \overline{w})^{s-1}}$$

Upshot: given $n \geq 3$ odd,

- ▶ There's a whole $n \times n$ matrix of higher Green's functions.
- ▶ Entries are 'coefficients' of $(n-1)$ -forms $g^{p,q}$ such that

$$dg^{p,q} = \nu^{p,q} + \overline{\nu^{q,p}}$$

where $\nu^{p,q}$ are algebraic n -forms which split the Hodge filtration on the weight $n+1$ part of $H^n(\overline{M}_{1,n} \setminus D_w)_{sgn}$.

- ▶ If $n = 2s - 1$, central entry is proportional to $G_s(z, w)$.
- ▶ Example:

$$\begin{pmatrix} G_{0,2}^{2,0}(z, w) & G_{0,2}^{1,0}(z, w) & G_{0,2}^{2,0}(z, w) \\ G_{1,1}^{2,0}(z, w) & G_{1,1}^{1,1}(z, w) & G_{1,1}^{0,2}(z, w) \\ G_{2,0}^{2,0}(z, w) & G_{0,2}^{1,1}(z, w) & G_{2,0}^{0,2}(z, w) \end{pmatrix}$$

with $G_{1,1}^{1,1}(z, w) \sim G_2(z, w)$.

Motives

- Let $K \subset \mathbb{C}$ be a number field. Many cohomology theories ‘with constant coefficients’ for algebraic varieties over K , e.g.

Betti	Alg. de Rham	ℓ -adic
$H_B^\bullet(X)$	$H_{dR}^\bullet(X)$	$H_\ell^\bullet(X)$
$H_{sing}^\bullet(X(\mathbb{C}); \mathbb{Q})$	$\mathbb{H}_{Zar}^\bullet(X; \Omega_{X/K}^\bullet)$	$\varprojlim_n H_{et}^\bullet(X; \mathbb{Z}/\ell^n) \otimes \mathbb{Q}_\ell$
$\mathbb{Q}$	K	$\mathbb{Q}_\ell$
complex conj.	Hodge filtration	Galois rep.

- Grothendieck comparison theorem; coefficients are **periods**:

$$H_{dR}^\bullet(X) \otimes_K \mathbb{C} \xrightarrow{\sim} H_B^\bullet(X) \otimes_{\mathbb{Q}} \mathbb{C}, \quad [\omega] \mapsto ([\sigma] \mapsto \int_\sigma \omega)$$

- The **motive** $H(X)$ is some ‘linear algebra object’ containing all possible cohomological information of X , for all Weil cohomology theories.

- ▶ $DM(K)$ = Voevodsky's **triangulated category** of motives over K . (Morally: bounded derived category of the 'true' abelian category of motives over K .)
- ▶ $DM(K)$ is $\mathbb{Q}$ -linear, triangulated, pseudo-abelian, rigid tensor.
- ▶ Functor:

$$Y \subset X \longmapsto \underbrace{H(X, Y)}_{\text{relative cohomology motive}} \in DM(K)$$

Denote $H(X, \emptyset) = H(X)$.

- ▶ Distinguished triangle:

$$H(Y)[-1] \longrightarrow H(X, Y) \longrightarrow H(X) \xrightarrow{+1}$$

“relative cohomology long exact sequence”

- Realisation functors:

$$R_B : DM(K) \rightarrow DVect_{\mathbb{Q}}, \quad R_{dR} : DM(K) \rightarrow DVect_K, \quad \dots$$

with $R_B H(X, Y) = H_B^{\bullet}(X, Y)$, ...

- Grothendieck: natural isomorphism $R_{dR} \otimes_K \mathbb{C} \cong R_B \otimes_{\mathbb{Q}} \mathbb{C}$.
- Basic motives:
 - Trivial: $\mathbb{Q}(0) = H(pt)$
 - Lefschetz: $\mathbb{Q}(-1) = H(\mathbb{A}^1 \setminus \{0\}, \{1\})[1]$.
 - Tate: $\mathbb{Q}(n) = \mathbb{Q}(-1)^{\otimes -n}$, $n \in \mathbb{Z}$.
- Tate twist: $M(n) = M \otimes \mathbb{Q}(n)$.
- Kummer motives:

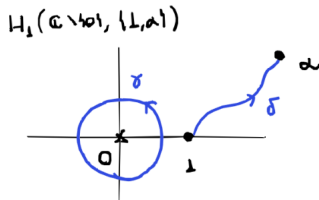
$$\mathcal{K}_{\alpha} = H(\mathbb{A}^1 \setminus \{0\}, \{1, \alpha\})[1].$$

Distinguished triangle:

$$\mathbb{Q}(0) \longrightarrow \mathcal{K}_{\alpha} \longrightarrow \mathbb{Q}(-1) \xrightarrow{+1}$$

Logarithms of algebraic numbers are periods of Kummer motives:

- $R_B \mathcal{K}_\alpha = H_B^1(\mathbb{A}^1 \setminus \{0\}, \{1, \alpha\})$ with $\mathbb{Q}$ -basis $\delta^\vee, \gamma^\vee$.



- $R_{dR} \mathcal{K}_\alpha = H_{dR}^1(\mathbb{A}^1 \setminus \{0\}, \{1, \alpha\})$ with K -basis $\frac{dz}{\alpha-1}, \frac{dz}{z}$.
- Period matrix (representing comparison isomorphism):

$$P = \begin{pmatrix} \int_\delta \frac{dz}{\alpha-1} & \int_\delta \frac{dz}{z} \\ \int_\gamma \frac{dz}{\alpha-1} & \int_\gamma \frac{dz}{z} \end{pmatrix} = \begin{pmatrix} 1 & \log \alpha \\ 0 & 2\pi i \end{pmatrix}$$

- Single-valued period matrix:

$$S = \overline{P}^{-1} P = \begin{pmatrix} 1 & \log |\alpha|^2 \\ 0 & -1 \end{pmatrix}$$

Every extension of $\mathbb{Q}(-1)$ by $\mathbb{Q}(0)$ in $DM(K)$ is generated by Kummer motives, namely:

$$\begin{aligned} K^\times \otimes_{\mathbb{Z}} \mathbb{Q} &\xrightarrow{\sim} Ext^1(\mathbb{Q}(-1), \mathbb{Q}(0)) \\ \alpha \otimes 1 &\mapsto \mathcal{K}_\alpha \end{aligned}$$

(By definition, $Ext^1(\mathbb{Q}(-1), \mathbb{Q}(0)) = Hom(\mathbb{Q}(-1), \mathbb{Q}(0)[1])$.)

Upshot:

To prove a number is a logarithm, write it as a period of some extension of $\mathbb{Q}(-1)$ by $\mathbb{Q}(0)$ in $DM(K)$, for some number field K .

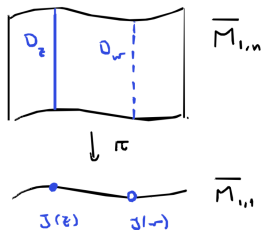
Putting everything together

► Motives:

$$M_{n,z,w} = H(\overline{M}_{1,n} \setminus D_w, D_z)_{\text{sgn}}[n], \quad F_{n,z} = H(D_z)_{\text{sgn}}[n-1]$$

► Biextension:

$$\begin{array}{c}
 H(\overline{M}_{1,n})_{\text{sgn}}[n] \\
 \downarrow \\
 F_{n,z} \longrightarrow M_{n,z,w} \longrightarrow H(\overline{M}_{1,n} \setminus D_w)_{\text{sgn}}[n] \xrightarrow{+1} \\
 \downarrow \\
 F_{n,w} \\
 \downarrow +1
 \end{array}$$

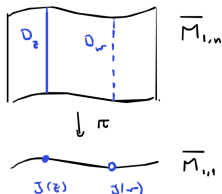


► If $H(\overline{M}_{1,n})_{\text{sgn}}[n] = 0$, we get a distinguished triangle:

$$F_{n,z} \longrightarrow M_{n,z,w} \longrightarrow F_{n,w}(-1) \xrightarrow{+1}$$

If $H(\overline{M}_{1,n})_{\text{sgn}}[n] = 0$, extension:

$$F_{n,z} \longrightarrow M_{n,z,w} \longrightarrow F_{n,w}(-1) \xrightarrow{+1}$$



Theorem (Brown-F.)

1. $G_{r,s}^{p,q}(z, w)$ are single-valued periods of $M_{n,z,w}$.
2. $F_{n,z} \cong \text{Sym}^{n-1} H^1(E_z)$. If z is CM, then $F_{n,z} = \mathbb{Q}(\frac{n-1}{2}) \oplus \tilde{F}_{n,z}$ after extending scalars.
3. If z, w are CM and $H(\overline{M}_{1,n})_{\text{sgn}}[n] = 0$, we get a subquotient

$$\mathbb{Q}(\frac{n-1}{2}) \longrightarrow \mathcal{GZ}_{n,z,w} \longrightarrow \mathbb{Q}(\frac{n-3}{2}) \xrightarrow{+1}$$

If $n = 2s - 1$, then $G_{s-1,s-1}^{s-1,s-1}(z, w)$ is 'the' single-valued period of $\mathcal{GZ}_{n,z,w}$. The *GZ conjecture holds* for $G_s(z, w)$.

We know that $R_B H(\overline{M}_{1,n})_{sgn}[n] = 0$ for $s \in \{1, 2, 3, 4, 5, 7\}$
(follows from Faber-Consani + classical theory of modular forms).

Conjecture

The functor $R_B : DM(K) \rightarrow DVect_{\mathbb{Q}}$ is conservative.

This proves “conservativity $\implies$ GZ”.

The vanishing of the above motives is not out of reach. For instance, conservativity is known to hold on the subcategory of **mixed Tate motives**.

Theorem (Brown-F.)

The motive of $\overline{M}_{1,3}$ is mixed Tate. In particular, the above argument proves the GZ conjecture for $n = 3$ (i.e., for $G_2(z, w)$).

We also expect the mixed Tate property for all $n \leq 10$ (compare with Belorousski’s rationality result).

Thank you!